ENGINEERING MATHEMATICS — B.TECH 1ST YEAR

Unit III
Partial Differential Equations

A Complete Step-by-Step Study Module

LEVEL
B.Tech Year 1

UNIT
Unit III – PDE

SEMESTER
1st Semester

40+ Solved Examples · 50 Exercise Problems · Complete Formula Sheet · Quick Revision Notes

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SECTION 01

Introduction to Partial Differential Equations

UnderstandingwhatPDEsare,howtheydifferfromODEs,and why they are important in engineering

1.1 What is a Differential Equation?

Before we understand **Partial Differential Equations (PDEs)**, let us first recall what a differential equation is.

> DEFINITION
>
> A **differential equation** is an equation that contains an unknown function and one or more of its *derivatives*. The equation describes a relationship between the function and how it changes.

For example, if a ball is thrown upward, its height `y` changes with time `t`. The equation describing this change involves the derivative `dy/dt` — that is a differential equation.

1.2 Ordinary vs Partial Differential Equations

Differential equations are classified based on the number of independent variables:

Feature	Ordinary DE (ODE)	Partial DE (PDE)
Independent variables	Only ONE (e.g., x or t)	TWO or MORE (e.g., x and y, or x, y, z)
Type of derivative	Ordinary derivatives (d/dx)	Partial derivatives ($\partial/\partial x$, $\partial\partial y$)
Example	$dy/dx + 2y = 0$	$\partial z/\partial x + \partial z/\partial y = 0$
Dependent variable	Function of ONE variable	Function of MULTIPLE variables
Notation	y', y'', dy/dx	p, q, r, s, t (standard PDE notation)

> DEFINITION — PARTIAL DIFFERENTIAL EQUATION
>
> A **Partial Differential Equation (PDE)** is an equation involving an unknown function of *two or more independent variables* and its *partial derivatives* with respect to those variables.

1.3 Standard Notation for PDEs

In engineering mathematics, when `z` is a function of two independent variables `x` and `y`, i.e., `z = f(x, y)`, the following shorthand notation is universally used:

Symbol	Meaning	Full Form
p	First partial derivative w.r.t. x	$\partial z/\partial x$
q	First partial derivative w.r.t. y	$\partial z/\partial y$
r	Second partial derivative w.r.t. x twice	$\partial^2 z/\partial x^2$
s	Mixed second partial derivative	$\partial^2 z/\partial x \partial y$
t	Second partial derivative w.r.t. y twice	$\partial^2 z/\partial y^2$

MEMORY TIP

Think of **p** and **q** as the "first floor" of derivatives (first order), and **r**, **s**, **t** as the "second floor" (second order). The letters go in alphabetical order: p, q → r, s, t.

1.4 Why Do Engineers Need PDEs?

PDEs appear naturally in almost every field of engineering because real-world problems often involve quantities that vary in multiple dimensions simultaneously:

Engineering Application	Governing PDE	Variables
Heat distribution in a rod/plate	Heat Equation: $\partial u/\partial t = \alpha\, \partial^2 u/\partial x^2$	Temperature u, position x, time t
Vibration of a string/membrane	Wave Equation: $\partial^2 u/\partial t^2 = c^2\, \partial^2 u/\partial x^2$	Displacement u, position x, time t
Electrostatics / Fluid flow	Laplace Equation: $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = 0$	Potential u, coordinates x, y
Fluid mechanics (Navier-Stokes)	Complex system of PDEs	Velocity, pressure, position, time

1.5 How to Identify a Partial Derivative

The key symbol that tells you something is a partial derivative is **∂** (called "del" or "partial"). This symbol means "differentiate with respect to this variable while keeping all other variables CONSTANT."

CORE CONCEPT — PARTIAL DIFFERENTIATION

If $z = x^2y + 3xy^2$, then:

∂z/∂x = differentiate treating y *as a constant* = $2xy + 3y^2$

∂z/∂y = differentiate treating x *as a constant* = $x^2 + 6xy$

SECTION 02

Order and Degree of PDE

LearningtoclassifyPDEsbytheirorderanddegree—fundamentalforchoosing the right solution method

2.1 Order of a PDE

DEFINITION — ORDER

The **order** of a PDE is the order of the *highest partial derivative* present in the equation.

To find the order, simply scan the equation for all partial derivatives and pick the one with the highest differentiation count.

PDE	Highest Derivative	Order
$\partial z/\partial x + \partial z/\partial y = 0$	$\partial z/\partial x$ or $\partial z/\partial y$ (both first order)	1st Order
$p + q = z$	$p = \partial z/\partial x$ (first order)	1st Order
$\partial^2 z/\partial x^2 + \partial^2 z/\partial y^2 = 0$	$\partial^2 z/\partial x^2$ (second order)	2nd Order
$r + 2s + t = 0$	$r = \partial^2 z/\partial x^2$ (second order)	2nd Order
$\partial^3 z/\partial x^3 + \partial z/\partial y = x$	$\partial^3 z/\partial x^3$ (third order)	3rd Order

2.2 Degree of a PDE

DEFINITION — DEGREE

The **degree** of a PDE is the *power (exponent)* of the highest order partial derivative, after the equation has been made free from radicals and fractions involving derivatives.

IMPORTANT CONDITION

The degree is defined ONLY when the PDE is a **polynomial** in its derivatives. If the equation contains expressions like sin(p) or e^q, the degree is NOT defined (or we say it is not a polynomial PDE).

PDE	Highest Order Derivative	Its Power	Degree
$p + q = 1$	p (1st order)	Power 1	1
$p^2 + q^2 = 1$	p (1st order)	Power 2	2
$(\partial^2 z/\partial x^2)^3 + \partial z/\partial y = 0$	$\partial^2 z/\partial x^2$ (2nd order)	Power 3	3
$\sqrt{p} + \sqrt{q} = 1$ (after squaring: $p + 2\sqrt{(pq)} + q = 1$)	p (1st order)	Needs rationalization	2 (after rationalization)
$r + 2s + t = 0$	r (2nd order)	Power 1	1

2.3 Linearity of a PDE

A PDE is called **linear** if the dependent variable (z) and all its partial derivatives appear in the *first degree only*, and no products of these appear.

PDE	Linear?	Reason
$p + q = z$	Yes — Linear	p, q, z all appear in degree 1; no products
$p \cdot q = z$	No — Non-linear	Product of p and q appears
$p^2 + q = x$	No — Non-linear	p^2 means p appears in degree 2
$x \cdot p + y \cdot q = z$	Yes — Linear	Coefficients are functions of x, y only
$p + q = z^2$	No — Non-linear	z^2 means dependent variable is in degree 2

KEY RULE FOR LINEARITY

A PDE is LINEAR if: (1) Every term containing z or its derivatives has degree exactly 1, AND (2) There are no products between z, p, q, r, s, t. The coefficients CAN be functions of x and y — that is perfectly fine.

2.4 Classification Summary

QUICK CLASSIFICATION REFERENCE

$$F(x, y, z, p, q, r, s, t, \ldots) = 0$$

Order = highest derivative present | Degree = power of that highest derivative |
Linear = all terms degree 1 in z and derivatives

SECTION 03

Formation of PDE by Elimination of Arbitrary Constants

How a relation with arbitrary constants gives rise to a PDE — the fundamental formation technique

3.1 Why Do We Form PDEs?

Consider a family of surfaces (or curves in 3D). Each surface is described by an equation involving arbitrary constants. When we **eliminate those constants**, we get a PDE that every member of the family satisfies.

CORE IDEA

If a relation $f(x, y, z, a, b) = 0$ involves **two arbitrary constants** *a* and *b*, then by differentiating partially with respect to x and y and eliminating *a* and *b*, we get a PDE in x, y, z, p, q.

3.2 Method of Elimination of Constants

The general procedure is:

1 **WRITE THE GIVEN RELATION**

Write the given function $z = f(x, y, a, b)$ clearly. Count the number of arbitrary constants (a, b).

2 **DIFFERENTIATE PARTIALLY W.R.T. X**

Differentiate the entire equation partially with respect to x, treating y as constant. This gives an equation involving $p = \partial z/\partial x$ and the constants.

3 **DIFFERENTIATE PARTIALLY W.R.T. Y**

Differentiate the original equation partially with respect to y, treating x as constant. This gives an equation involving $q = \partial z/\partial y$ and the constants.

4

ELIMINATE THE CONSTANTS

> Using the three equations (original + two differentiated), eliminate the arbitrary constants a and b algebraically. The result is the PDE.

Example 1 Form PDE from z = ax + by + ab

Given: $z = ax + by + ab$

1 IDENTIFY THE RELATION

We have: $z = ax + by + ab$

Here, *z* is the dependent variable (a function of x and y).

The arbitrary constants are **a** and **b**.

Our goal: eliminate both **a** and **b** to form a PDE.

2 DIFFERENTIATE PARTIALLY W.R.T. X

We differentiate $z = ax + by + ab$ with respect to x, treating y (and hence b) as constants:

$$\partial z/\partial x = a\cdot(1) + b\cdot(0) + 0 = a$$

Therefore: $p = a$

Explanation: The derivative of ax w.r.t. x is a. The term by does not contain x, so its derivative is 0. The constant ab has no x, so its derivative is also 0.

3 DIFFERENTIATE PARTIALLY W.R.T. Y

We differentiate $z = ax + by + ab$ with respect to y, treating x (and hence a) as constants:

$$\partial z/\partial y = a\cdot(0) + b\cdot(1) + 0 = b$$

Therefore: $q = b$

Explanation: The term ax has no y, so its derivative is 0. The derivative of by w.r.t. y is b. The constant ab has no y, so its derivative is 0.

4 ELIMINATE CONSTANTS A AND B

From Steps 2 and 3, we found: $a = p$ and b = q

Now substitute a = p and $b = q$ into the original equation:

$$z = (p)\cdot x + (q)\cdot y + (p)\cdot(q)$$

We simply replaced a with p and b with q. No more arbitrary constants remain!

5 WRITE THE FINAL PDE

$$z = px + qy + pq$$

This is a first-order PDE in the standard form.

FINAL ANSWER

$$z = px + qy + pq$$

CONCEPT SUMMARY

When a relation involves *n* arbitrary constants, we need to differentiate *n* times (partially, in different directions) and then eliminate the constants. Here, 2 constants needed 2 differentiations.

Example 2 Form PDE from $z = (x^2 + a)(y^2 + b)$

$$\text{Given: } z = (x^2 + a)(y^2 + b)$$

1 IDENTIFY CONSTANTS

The arbitrary constants are **a** and **b**. We need to eliminate both.

2 DIFFERENTIATE W.R.T. X

$$p = \partial z/\partial x = (2x)(y^2 + b)$$

Using product rule (but $y^2 + b$ is constant w.r.t. x), so: $d/dx[(x^2 + a)] = 2x$, and $(y^2 + b)$ stays as is.

So: $p = 2x(y^2 + b)$, which gives us $y^2 + b = p/(2x)$

3 DIFFERENTIATE W.R.T. Y

$$q = \partial z/\partial y = (x^2 + a)(2y)$$

Here, $(x^2 + a)$ is constant w.r.t. y, and $d/dy[(y^2 + b)] = 2y$.

So: $q = 2y(x^2 + a)$, which gives us $x^2 + a = q/(2y)$

4 ELIMINATE CONSTANTS

From steps above: $y^2 + b = p/(2x)$ and $x^2 + a = q/(2y)$

Substitute back into original equation $z = (x^2 + a)(y^2 + b)$:

$$z = [q/(2y)] \cdot [p/(2x)]$$

$$z = pq / (4xy)$$

Therefore:

$$4xyz = pq$$

FINAL ANSWER

$$4xyz = pq \quad \text{or} \quad pq = 4xyz$$

Example 3 Form PDE from $z = ax^2 + by^2$

Given: $z = ax^2 + by^2$

1 DIFFERENTIATE PARTIALLY W.R.T. X

$$p = \partial z/\partial x = 2ax + 0 = 2ax$$

Therefore: $a = p/(2x)$

Treating y as constant, $d/dx[ax^2] = 2ax$, and $d/dx[by^2] = 0$.

2 DIFFERENTIATE PARTIALLY W.R.T. Y

$$q = \partial z/\partial y = 0 + 2by = 2by$$

Therefore: $b = q/(2y)$

Treating x as constant, $d/dy[ax^2] = 0$, and $d/dy[by^2] = 2by$.

3 SUBSTITUTE BACK

Substitute $a = p/(2x)$ and $b = q/(2y)$ into $z = ax^2 + by^2$:

$$z = [p/(2x)] \cdot x^2 + [q/(2y)] \cdot y^2$$

$$z = px/2 + qy/2$$

Multiply both sides by 2:

$$2z = px + qy$$

FINAL ANSWER

$$px + qy = 2z$$

SECTION 04

Formation of PDE by Elimination of Arbitrary Functions

A more powerful formation technique — eliminating arbitrary functions using partial differentiation

4.1 The Concept

Sometimes a relation involves an **arbitrary function** (like φ, f, F, ψ) rather than just constants. For example:

$$z = \varphi(x^2 + y^2) \qquad \text{or} \qquad z = f(y/x) \qquad \text{or} \qquad F(x + y, x - z) = 0$$

Since these involve arbitrary functions (infinitely many possibilities), we need to differentiate and *eliminate the function* itself to get a PDE.

CORE PRINCIPLE

If a relation involves **one arbitrary function**, one differentiation (by the chain rule) is enough to eliminate it — yielding a **first-order** PDE. If two arbitrary functions are present, two differentiations are needed, yielding a **second-order** PDE.

4.2 Method: One Arbitrary Function

When the relation is of the form $z = f(u)$ where u is some expression in x and y:

1. **WRITE Z = F(U) AND FIND ∂U/∂X AND ∂U/∂Y**

   Identify what u is (e.g., $u = x^2 + y^2$, $u = y/x$, etc.) and compute its partial derivatives.

2. **APPLY CHAIN RULE**

$$p = \partial z/\partial x = f'(u) \cdot \partial u/\partial x$$

$$q = \partial z/\partial y = f'(u) \cdot \partial u/\partial y$$

3 ELIMINATE F'(U)

> Divide one equation by the other to eliminate the unknown f'(u), or manipulate algebraically to get a PDE in p and q.

Example 4 Form PDE from $z = f(x^2 + y^2)$

Given: $z = f(x^2 + y^2)$

1 IDENTIFY THE STRUCTURE

Let $u = x^2 + y^2$, so $z = f(u)$

The arbitrary function is *f* — we don't know what function it is, but we can still eliminate it!

2 COMPUTE PARTIAL DERIVATIVES OF U

$\partial u/\partial x = 2x$ (derivative of $x^2 + y^2$ w.r.t. x, treating y constant)

$\partial u/\partial y = 2y$ (derivative of $x^2 + y^2$ w.r.t. y, treating x constant)

3 APPLY CHAIN RULE TO FIND P AND Q

Using $p = \partial z/\partial x = f'(u) \cdot \partial u/\partial x$:

$$p = f'(u) \cdot 2x \quad \rightarrow \quad f'(u) = p/(2x)$$

Using $q = \partial z/\partial y = f'(u) \cdot \partial u/\partial y$:

$$q = f'(u) \cdot 2y \quad \rightarrow \quad f'(u) = q/(2y)$$

4 ELIMINATE F'(U)

Since both expressions equal f'(u), set them equal:

$$p/(2x) = q/(2y)$$

Cross multiply:

$$2y \cdot p = 2x \cdot q$$

$$py = qx$$

FINAL ANSWER

$$py = qx \qquad \text{or} \qquad py - qx = 0$$

Example 5 Form PDE from $z = f(y/x)$

Given: $z = f(y/x)$

1 LET U = Y/X

We have $u = y/x$, so $z = f(u)$.

Compute partial derivatives of u:

$$\partial u/\partial x = -y/x^2 \qquad \text{(using quotient rule: } d/dx[y/x] = -y/x^2\text{)}$$

$$\partial u/\partial y = 1/x \qquad \text{(derivative of } y/x \text{ w.r.t. } y = 1/x\text{)}$$

2 FIND P AND Q BY CHAIN RULE

$$p = f'(u) \cdot (-y/x^2) \quad \rightarrow \quad f'(u) = -px^2/y$$

$$q = f'(u) \cdot (1/x) \quad \rightarrow \quad f'(u) = qx$$

3 EQUATE AND SIMPLIFY

Setting both expressions for $f'(u)$ equal:

$$-px^2/y = qx$$

Multiply both sides by y:

$$-px^2 = qxy$$

Divide by x (assuming $x \neq 0$):

$$-px = qy$$

Rearrange:

$$px + qy = 0$$

FINAL ANSWER

$$px + qy = 0$$

Example 6 Form PDE from $z = f(x) + g(y)$

Given: $z = f(x) + g(y)$ — two arbitrary functions

1 **FIRST PARTIAL DERIVATIVES**

$$p = \partial z/\partial x = f'(x) + 0 = f'(x)$$

$$q = \partial z/\partial y = 0 + g'(y) = g'(y)$$

f depends only on x, so $\partial f/\partial y = 0$. g depends only on y, so $\partial g/\partial x = 0$.

2 **SECOND MIXED PARTIAL DERIVATIVE**

Differentiate $p = f'(x)$ with respect to y:

$$\partial p/\partial y = \partial/\partial y[f'(x)] = 0 \qquad \text{(since } f'(x) \text{ has no } y\text{)}$$

But we know: $\partial p/\partial y = \partial^2 z/\partial y\partial x = s$

$$s = 0$$

FINAL ANSWER

$$s = 0 \qquad \text{(i.e.,} \quad \partial^2 z/\partial x\partial y = 0)$$

KEY INSIGHT

Two arbitrary functions require second-order differentiation. The resulting PDE is second order.

First Order Linear PDE

Understandingfirst-orderPDEsandhowtorecognizetheirstructure before applying solution methods

5.1 Definition of First Order Linear PDE

DEFINITION

A **first-order linear PDE** is a PDE in which the partial derivatives p (= ∂z/∂x) and q (= ∂z/∂y) appear in the *first degree only*, with coefficients that are functions of x, y, z only (not of p and q).

5.2 The Standard Form

The most general first-order linear PDE in z = z(x, y) is:

STANDARD FORM OF FIRST-ORDER LINEAR PDE (LAGRANGE'S FORM)

$$Pp + Qq = R$$

where P, Q, R are functions of x, y, z only (not of p or q)

$p = \partial z/\partial x \quad q = \partial z/\partial y$

5.3 What P, Q, R Represent

Symbol	Meaning	Example (Pp + Qq = R)
P	Coefficient of p = ∂z/∂x	In xp + yq = z: P = x
Q	Coefficient of q = ∂z/∂y	In xp + yq = z: Q = y
R	Right-hand side (function of x,y,z)	In xp + yq = z: R = z

5.4 Examples of First-Order Linear PDEs

PDE	P	Q	R	Type
$p + q = 1$	1	1	1	Constant coefficients
$xp + yq = z$	x	y	z	Variable coefficients
$yp - xq = 0$	y	−x	0	Homogeneous RHS
$p + xq = y$	1	x	y	Mixed coefficients
$(y-z)p + (z-x)q = x-y$	y−z	z−x	x−y	Classic Lagrange form

WHY IS IT CALLED "LINEAR"?

The equation $Pp + Qq = R$ is linear because p and q each appear with power 1. There is no p^2, no q^2, no pq. The coefficients P, Q, R can be ANY function of x, y, z — that does not affect linearity.

5.5 Non-linear First-Order PDEs (for comparison)

The following are NOT linear (for awareness):

PDE	Why Non-Linear
$p^2 + q^2 = 1$	p and q appear with power 2
$pq = z$	Product of p and q (degree 2)
$p + q^2 = xy$	q appears with power 2

SECTION 06

Standard Form of PDE

Recognizing and converting PDEs into standard solvable forms

6.1 Lagrange's Linear Equation — The Standard Form

LAGRANGE'S STANDARD FORM

A first-order linear PDE of the form **Pp + Qq = R** is called **Lagrange's linear equation**, where P, Q, R are functions of x, y, z. This is the most important form in first-order PDE theory.

6.2 Converting to Lagrange's Form

Before solving, we must convert any given PDE into the form `Pp + Qq = R`. Let us see how:

Original PDE	Rearranged to Pp + Qq = R	P, Q, R
$z \cdot \partial z/\partial x + z \cdot \partial z/\partial y = x + y$	$z \cdot p + z \cdot q = x + y$	P=z, Q=z, R=x+y
$\partial z/\partial x - 2\partial z/\partial y = 3z$	$p - 2q = 3z \rightarrow (1)p + (-2)q = 3z$	P=1, Q=−2, R=3z
$y^2p - xyq = x(z - 2y)$	Already in form Pp + Qq = R	$P=y^2$, $Q=-xy$, $R=x(z-2y)$

6.3 Standard Forms Classification

Form Name	Equation Pattern	Solution Approach
Lagrange's Form	Pp + Qq = R	Auxiliary equations
Charpit's Form	F(x, y, z, p, q) = 0 (non-linear)	Charpit's method
Homogeneous Form	All terms same degree in D, D'	CF + PI method
Non-homogeneous Form	Mixed degree terms	Special methods

EXAM FOCUS

For B.Tech 1st Year, the PRIMARY focus is on **Lagrange's form: Pp + Qq = R**. Master this completely first!

SECTION 07

Lagrange's Method — Concept & Theory

The complete theoretical foundation of Lagrange's method, explained from first principles

7.1 Who Was Lagrange?

Joseph-Louis Lagrange (1736–1813) was an Italian-French mathematician who made fundamental contributions to analysis, number theory, and mechanics. The method we use to solve first-order linear PDEs is named after him.

7.2 The Problem We Are Solving

We want to find `z = z(x, y)` — a surface in three dimensions — that satisfies:

```
Pp + Qq = R       where P = P(x,y,z), Q = Q(x,y,z), R = R(x,y,z)
```

7.3 The Geometric Interpretation

The equation Pp + Qq = R has a beautiful geometric meaning:

GEOMETRIC MEANING

The equation **Pp + Qq = R** says that the *normal to the solution surface* (−p, −q, 1) is *perpendicular to the vector* (P, Q, R). This means the vector (P, Q, R) lies in the tangent plane of the solution surface at every point. This is why we look for *integral curves* of the system dx/P = dy/Q = dz/R — these curves lie on the solution surface!

7.4 The Main Theorem — Lagrange's Theorem

LAGRANGE'S THEOREM

The **general solution** of the first-order linear PDE `Pp + Qq = R` is:

F(u, v) = 0 or equivalently **u = f(v)** or **v = g(u)**

where $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ are two independent solutions (first integrals) of the **auxiliary**

equations (also called subsidiary equations or characteristic equations):

dx/P = dy/Q = dz/R

and F is an arbitrary function.

7.5 Why Does This Work?

This is the deep reason: if $u = c_1$ and $v = c_2$ are solutions of the auxiliary equations, then any function $F(u, v) = 0$ also satisfies the original PDE $Pp + Qq = R$. The proof involves showing that:

VERIFICATION (CONCEPTUAL)

If $u = c_1$ satisfies dx/P = dy/Q = dz/R, then:

$P\cdot(\partial u/\partial x) + Q\cdot(\partial u/\partial y) + R\cdot(\partial u/\partial z) = 0$ (This is the PDE of first integrals)

Similarly for $v = c_2$.

Then F(u,v) = 0 can be differentiated with respect to x and y to get exactly Pp + Qq = R.

7.6 Steps of Lagrange's Method (Summary)

Step	Action	Result
Step 1	Write PDE in form Pp + Qq = R	Identify P, Q, R
Step 2	Write auxiliary equations dx/P = dy/Q = dz/R	Set up the system
Step 3	Solve dx/P = dy/Q to get $u(x,y,z) = c_1$	First integral
Step 4	Solve dy/Q = dz/R (or dx/P = dz/R) to get $v(x,y,z) = c_2$	Second integral
Step 5	Write general solution F(u, v) = 0 or u = f(v)	General solution

COMMON MISTAKE

Students often write down the auxiliary equations incorrectly. Remember: the denominator of dx is P (coefficient of p), the denominator of dy is Q (coefficient of q), and the denominator of dz is R (the right-hand side). Do NOT mix these up!

SECTION 08

Auxiliary Equations — Deep Understanding

Mastering the subsidiary equations dx/P = dy/Q = dz/R and techniques to solve them

8.1 What Are Auxiliary Equations?

DEFINITION — AUXILIARY (SUBSIDIARY) EQUATIONS

For the PDE $Pp + Qq = R$, the associated system of ordinary differential equations:

$$dx/P = dy/Q = dz/R$$

is called the **auxiliary equations** (also: subsidiary equations, Lagrange-Charpit equations, or characteristic equations). Solving this system gives the two independent solutions (first integrals) needed for the general solution.

8.2 How to Solve the Auxiliary Equations

The auxiliary system $dx/P = dy/Q = dz/R$ is a system of simultaneous ODEs. We solve it by taking pairs:

Pair	Equation	What it gives
Pair 1	$dx/P = dy/Q$	Relation between x and y→ $u(x,y) = c_1$
Pair 2	$dy/Q = dz/R$	Relation between y and z→ $v(y,z) = c_2$
Pair 3	$dx/P = dz/R$	Relation between x and z→ can be used instead of pair 2

8.3 Techniques for Solving the System

There are several clever techniques depending on the form of P, Q, R:

Technique 1: Direct Integration (when variables separate)

If the pair dx/P = dy/Q gives a directly integrable ODE, integrate both sides.

Example: $dx/x = dy/y$ → $\ln x = \ln y + C$ → $x/y = c_1$

Technique 2: Grouping / Multiplier Method

If $dx/P = dy/Q = dz/R = (l\cdot dx + m\cdot dy + n\cdot dz)/(l\cdot P + m\cdot Q + n\cdot R)$, then choose l, m, n so the denominator simplifies to 0 or an exact expression. This is the **method of multipliers**.

> **METHOD OF MULTIPLIERS**
>
> Using the property of equal ratios: If $dx/P = dy/Q = dz/R$, then for any constants l, m, n:
>
> $$dx/P = dy/Q = dz/R = (l\cdot dx + m\cdot dy + n\cdot dz)/(l\cdot P + m\cdot Q + n\cdot R)$$
>
> Choose l, m, n so that **$l\cdot P + m\cdot Q + n\cdot R = 0$**. Then the numerator $l\cdot dx + m\cdot dy + n\cdot dz$ is an **exact differential** equal to zero, which can be integrated directly.

8.4 Important Property of Equal Ratios

If $a/b = c/d$, then also $a/b=c/d=(a+c)/(b+d)$. This extends to: if each ratio equals k, then $(l\cdot a + m\cdot c)/(l\cdot b + m\cdot d) = k$ for any l, m. This is the mathematical basis for the multiplier method.

8.5 When Can You Use Multipliers?

Situation	Good Multipliers to Try
$P + Q + R = 0$ is identically satisfied	$l = m = n = 1$ (add all three)
x, y, z appear symmetrically	Try l=x, m=y, n=z
$x\cdot P + y\cdot Q + z\cdot R$ simplifies nicely	l=x, m=y, n=z
One pair is separable	Solve that pair directly first